

B.Tech Computer Science and Engineering (Cyber Security) /B.Tech(Cyber Security)
KURUKSHETRAUNIVERSITY, KURUKSHETRA

SCHEME OF EXAMS W.E.F THE SESSION 2026-27

SEMESTER-V

S. No.	Course No./ Code	Subject	L:T:P	Hours/ Week	Credits	Examination Schedule (Marks)				Duration of Exam (Hours)
						End Semester Exam	Internal assessment	Practical Exam	Total	
1	B24-CS-CYS-301	Computer Network	3:1:0	4	4	70	30	0	100	3
2	B24-CS-CYS-303	Database Security	3:1:0	4	4	70	30	0	100	3
3	B24-CS-CYS-305	Cryptography	3:1:0	4	4	70	30	0	100	3
4	B24-BSC-CYS-301	Probability Theory	3:0:0	3	3	70	30	0	100	3
5	B24-CS-CYS-307	Information Security and Data Hiding	3:0:0	3	3	70	30	0	100	3
6	---	Open Elective-I	3:0:0	3	3	70	30	0	100	3
7	B24-CS-CYS-301L	Computer Network Lab	0:0:3	3	1.5	0	40	60	100	3
8	B24-CS-CYS-303L	Database Security Lab	0:0:3	3	1.5	0	40	60	100	3
9	B24-CS-CYS-305L	Cryptography lab	0:0:2	2	1	0	40	60	100	3
10	SIM-301*	Summer Internship	2:0:0	2	1	--	100	0	100	3
TOTAL				31	26	420	400	180	1000	

Open Elective-I	
Soft Skills and Interpersonal Communication : B24-CS-CYS-OE-301	
Innovation and Entrepreneurship : B24-CS-CYS-OE-303	
Enterprise Resource Planning : B24-CS-CYS-OE-305	

Note: SIM-301 is a mandatory credit-less course in which the students will be evaluated for the Summer Internship undergone after 4th semester and students will be required to get passing marks to qualify.

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SCHEME OF EXAMS W.E.F THE SESSION 2026-27

SEMESTER-VI

S. No.	Course No./ Code	Subject	L:T:P	Hours/ Week	Credits	Examination Schedule(Marks)				Duration of Exam(Hours)
						End Semester Exam	Internal assessment	Practical Exam	Total	
1	B24-CS-CYS-302	Operating System	3:1:0	4	4	70	30	0	100	3
2	B24-CS-CYS-304	Digital Forensics	3:1:0	4	4	70	30	0	100	3
3	B24-CS-CYS-306	Ethical Hacking	3:1:0	4	4	70	30	0	100	3
4	B24-CS-CYS-308	Cloud Computing	3:1:0	4	4	70	30	0	100	3
5	B24-BSC-CYS-302	Linear Algebra and Statistics	3:0:0	3	3	70	30	0	100	3
6	B24-CS-CYS-306L	Ethical Hacking Lab	0:0:3	3	1.5	0	40	60	100	3
7	B24-CS-CYS-308L	Cloud Computing Lab	0:0:3	3	1.5	0	40	60	100	3
8	B24-CS-CYS-304L	Digital Forensics Lab	0:0:2	2	1	0	40	60	100	3
9	---	Open Elective-II	3:0:0	3	3	70	30	0	100	3
TOTAL				30	26	420	300	180	900	

Open Elective-II	
Law of Privacy : B24-CS-CYS-OE-302	
Reinforcement Learning : B24-CS-CYS-OE-304	
Research methodology : B24-CS-CYS-OE-306	

B24-CS-CYS-301	Computer Network						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3 Hours
Purpose	To introduce the architecture and layers of computer network, protocols used at different layers.						
Course Outcomes (CO)							
CO1	To understand the basic concept of networking, types, networking topologies and layered architecture.						
CO2	To understand data link layer and MAC sub-layer`						
CO3	To understand the network Layer functioning						
CO4	To understand the transport layer and application layer operation						

Unit -1

Introduction: Introduction to Computer Networks, Data Communication System and its components, Data Flow. Wireless Vs Wired networks, broadcast VS point-to-point networks, Network topologies, protocols, interfaces and services, ISO-OSI reference model Vs TCP/IP architecture.

Physical Layer: Concept of Analog Vs Digital Signal, Bandwidth. Multiplexing : Frequency Division, Time Division, Wavelength Division,

Introduction to Transmission Media : Twisted pair, Coaxial cable, Fiber optics, Wireless transmission (radio, microwave, infrared), Switching: Circuit Switching, Message Switching ,Packet Switching & comparisons, narrowband ISDN, broadband ISDN and ATM.

Unit -2

Data link layer: Error Control, Types of errors, framing(character and bit stuffing), error detection & correction methods; Flow control; Protocols: Stop & wait ARQ, Go-Back- N ARQ, sliding window protocols, Selective repeat ARQ, HDLC

Medium access sub layer: Point to point protocol, FDDI, token bus, token ring; Reservation, polling, Multiple access protocols: Pure ALOHA, Slotted ALOHA, CSMA, CSMA/CD, FDMA, TDMA, CDMA, LLC, Traditional Ethernet, fast Ethernet, Network devices-repeaters, hubs, switches, Bridges, Router, Gateway .

Unit-3

Network layer: Addressing : Internet address, subnetting; Routing techniques, static vs. dynamic routing , routing table, DHCP, IEEE standards 802.x, Routing algorithms: shortest path algorithm, flooding, distance vector routing, link state routing; Protocols: ARP, RARP, IP, ICMP, IGMP, IPV6; Unicast and multicast routing protocols.

Unit-4

Transport layer: Process to process delivery; UDP; TCP, RPC, Congestion control algorithm: Leaky bucket algorithm, Token bucket algorithm, choke packets; Quality of service: techniques to improve QoS.

Application layer: DNS; SMTP, SNMP, FTP, HTTP & WWW; Firewalls, Bluetooth, Email, S/MIME, IMAP,

Security: Cryptography, user authentication, security protocols in internet, public key encryption algorithm, digital signatures.

TEXT BOOK

1. Behrouz A. Forouzan, “Data communication and Networking”, Tata McGraw Hill, Fourth Edition, 2011.
2. Computer Networks, 4th Edition, Pearson Education by Andrew S. Tanenbaum

REFERENCES

1. Larry L.Peterson, Peter S. Davie, “Computer Networks”, Elsevier, Fifth Edition, 2012.
2. William Stallings, “Data and Computer Communication”, Eighth Edition,Pearson Education, 2007.

3. James F. Kurose, Keith W. Ross, “Computer Networking: A Top–Down Approach Featuring the Internet”, Pearson Education, 2005.

B24-CS-CYS-303	Database Security						
Lecture	Tutorial	Practical	Credits	Internal Assessment	End Semester Exam	Total Marks	Time
3	1	0	4	30	70	100	3hour
Purpose: to understand how to protect digital assets from unauthorized access, misuse, and theft							
Course Outcomes							
CO1	Understand various authentication techniques and identify common design and implementation flaws.						
CO2	To gain proficiency in reviewing source code to identify security vulnerabilities in various platforms (Java, ASP.NET, PHP, Perl, Javascript, SQL) and learn about fundamental data security requirements						
CO3	Understand the mechanisms of user authentication and password protection in databases.						
CO4	Understand how to protect the data dictionary, manage system and object privileges, and secure data outsourcing						

Unit I

Introduction To Database Security: Fundamental Data Security Requirements, Data Security Concerns, Compliance Mandates, Security Risks, Developing Enterprise Security Policy, Defining a Security Policy, Implementing a Security Policy, Techniques to Enforce Security

Unit II

Vulnerability Analysis of Source Code: Approaches to Code Review, Signatures of Common Vulnerabilities, Analysis of Java platform, Analysis of ASP.NET platform, Analysis of PHP, Analysis of Perl, Analysis of Javascript, Analysis of SQL

Unit III

Database Access Control: User Authentication, Protecting Passwords, Creating Fixed Database Links, Encrypting Database Link Passwords, Using Database Links Without Credentials, Using Database Links and Changing Passwords, Auditing with Database Links, Restricting A Database Link with Views, Trust Management & Negotiation

Unit IV

Database Security Issues: Database Security Basics, Security Checklist, Reducing Administrative Effort, Applying Security Patches, Default Security Settings, Secure Password Support, Enforcing Password Management, Protecting The Data Dictionary, System and ObjectPrivileges, Secure Data Outsourcing, Security in Advanced Database Systems, Managing Enterprise User Security.

SUGGESTED BOOKS

- “ Database Security” ,S.Castano, M. Fugini, G. Martella,P.Samarati, Addison-Wesley
- “ Database Security “ Alfred Basta, Melissa Zgola, Cengage Publication, 2012.

B24-CS-CYS-305	Cryptography						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3 Hours
Purpose	To understand the importance of cryptanalysis in our increasingly computer-driven world.						
Course Outcomes (CO)							
CO1	To develop a mathematical foundation for the study of cryptography.						
CO2	To understand Number Theory and Algebra for design of cryptographic algorithms						
CO3	To understand the role of cryptography in communication over an insecure channel.						
CO4	To analyze the different public key encryption schemesand elliptic curves andpairings based on different security models						

UNIT-I

Classical cryptography and overview: Introduction to Cryptographic Techniques, Classical cryptosystems and their cryptanalysis, Model of secure communication, Security services, Overview of attacks, X.800 Security Architecture for Open System Interconnection (OSI), and cryptanalysis. Security Protocols: Authentication: Kerberos, X.509 Authentication Service, Secure Socket Layer (SSL), Secure Electronic Transaction (SET), Email security, PGP, S/MIME, IP Security (IPSec)

UNIT II

Mathematical background: Introduction to Number theory, Modular arithmetic, prime number generation, GCD, Euclidean Algorithm, Extended Euclidean Algorithm, Fast Exponentiation, Chinese Remainder Theorem, Fermat's and Euler's Theorem

UNIT III

Private key cryptography: Symmetric Encryption. Definitions. Chosen Plaintext Attack. Chosen Ciphertext Attack, Variations on DES, RC4, RC5 and Blowfish, Hash Functions. Message Authentication Codes. Collision-Resistance. Public key cryptography: RSA, ElGamal, DSA, Elliptic curve cryptosystems, Public Key Cryptography standard (PKCS), PKI, Digital Certificates, and Key management techniques, message digest: SHA-1, MD5

UNIT IV

Public Key Cryptosystems of RSA, Rabin, etc. their security and cryptanalysis. Finite fields: Construction and examples. Diffie-Hellman key exchange. Discrete logarithm problem in general and on finite fields. Cryptosystems based on Discrete logarithm problems such as Massey-Omura cryptosystems. Algorithms for finding discrete logarithms, their analysis. Polynomials on finite fields and their factorization/irreducibility and their application to coding theory. Elliptic curves, Public key cryptosystems particularly on Elliptic curves. Problems of key exchange, discrete logarithms and the elliptic curve logarithm problem. Implementation of elliptic curve cryptosystems

Suggested Books:

1. Behrouz A. Forouzan and D. Mukhopadhyay, Cryptography & Network Security, McGraw Hill, New Delhi.
2. William Stallings, Cryptography and Network Security: Principles and Practice, Prentice Hall
3. Douglas R. Stinson, Cryptography: Theory and Practice, Chapman and Hall
4. J. Katz and Y. Lindell, Introduction to Modern Cryptography, CRC press

B24-BSC-CYS-301		Probability Theory					
L	T	P	Credit	End Semester Exam	Internal Assessment	Total	Time
3	0	0	3	70	30	100	3 Hrs.
Purpose	To develop understanding of probability theory, random variables, and stochastic processes for analyzing uncertainty and modeling random phenomena in engineering and science.						
CO 1	Understand and apply fundamental concepts of probability, random variables, and conditional probability to analyze random experiments.						
CO 2	Analyze probability distribution functions and compute statistical measures such as expectation, moments, variance, and transformations of random variables.						
CO3	Evaluate joint distributions and statistical relationships of multiple random variables and apply the central limit theorem.						
CO4	Understand stochastic processes and analyze temporal characteristics such as stationarity, correlation, and system response.						

UNIT I

Probability and Random Variable Probability: Set theory, Experiments and Sample Spaces, Discrete and Continuous Sample Spaces, Events, Probability Definitions and Axioms, Mathematical Model of Experiments, Joint Probability, Conditional Probability, Total Probability, Bayes' Theorem, and Independent Events, Bernoulli's trials. The Random Variable: Definition of a Random Variable, Conditions for a Function to be a Random Variable, Discrete and Continuous, Mixed Random Variable

UNIT II

Distribution and density functions and Operations on One Random Variable Distribution and density functions: Distribution and Density functions, Properties, Binomial, Poisson, Uniform, Exponential Gaussian, Rayleigh and Conditional Distribution, Methods of defining Conditioning Event, Conditional Density function and its properties, problems. Operation on One Random Variable: Expected value of a random variable, function of a random variable, moments about the origin, central moments, variance and skew, characteristic function, moment generating function, transformations of a random variable, monotonic transformations for a continuous random variable, non monotonic transformations of continuous random variable, transformations of Discrete random variable

UNIT III

Multiple Random Variables and Operations on Multiple Random Variables Multiple Random Variables: Vector Random Variables, Joint Distribution Function and Properties, Joint density Function and Properties, Marginal Distribution and density Functions, conditional Distribution and density Functions, Statistical Independence, Distribution and density functions of Sum of Two Random Variables and Sum of Several Random Variables, Central Limit Theorem - Unequal Distribution, Equal Distributions Operations on Multiple Random Variables: Expected Value of a Function of Random Variables, Joint Moments about the Origin, Joint Central Moments, Joint Characteristic Functions, and Jointly Gaussian Random Variables: Two Random Variables case and N Random Variable case, Properties, Transformations of Multiple Random Variables

UNIT IV

Stochastic Processes-Temporal Characteristics: The Stochastic process Concept, Classification of Processes, Deterministic and Nondeterministic Processes, Distribution and Density Functions, Statistical Independence and concept of Stationarity: First-Order Stationary Processes, Second-Order and Wide-Sense Stationarity, Nth-Order and Strict-Sense Stationarity, Time Averages and Ergodicity, Mean-Ergodic Processes, Correlation-Ergodic Processes Autocorrelation Function and Its Properties, Cross-Correlation Function and Its Properties, Covariance Functions and its properties, Gaussian Random Processes

TEXT BOOKS:

1. Probability, Random Variables & Random Signal Principles -Peyton Z. Peebles, TMH, 4th Edition, 2001.
2. Probability and Random Processes-Scott Miller, Donald Childers,2Ed,Elsevier,2012

REFERENCE BOOKS:

1. Theory of probability and Stochastic Processes-Pradip Kumar Gosh, University Press
2. Probability and Random Processes with Application to Signal Processing - Henry Stark and John W. Woods, Pearson Education, 3rd Edition.
3. Probability Methods of Signal and System Analysis- George R. Cooper, Clave D. MC Gillem, Oxford, 3rd Edition, 1999.
4. Statistical Theory of Communication -S.P. Eugene Xavier, New Age Publications 2003
5. Probability, Random Variables and Stochastic Processes Athanasios Papoulis and S.Unnikrishna Pillai, PHI, 4th Edition, 2002

B24-CS-CYS-307	Information Security and Data Hiding						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	0	0	3	70	30	100	3Hrs.
Purpose	To understand and apply techniques for securing digital information and communication systems, and to implement data hiding methods to protect confidentiality, integrity, and ownership of digital data.						
Course Outcomes(CO)							
CO1	To understand different types of security attacks and security services, and apply cryptographic techniques such as classical encryption, DES, and Blowfish along with key distribution and random number generation to ensure secure communication in networks.						
CO2	To understand and apply public key cryptography, key management, and secure communication protocols, implement message authentication using hash functions and MACs, and analyze IP security mechanisms (IPsec, AH, ESP) to ensure confidentiality, integrity, and authentication in networks.						
CO3	To understand and apply data hiding techniques including steganography and digital watermarking, analyze security and privacy aspects of multimedia data, classify and implement various steganography algorithms, and develop solutions for applications in covert communication, e-commerce, military, and digital content protection.						
CO4	To understand digital rights management (DRM) and copyright protection techniques, implement authentication and digital signature methods, analyze multimedia security and information assurance, evaluate attacks and performance of data hiding systems, and apply data hiding technologies in practical domains such as medicine, law enforcement, remote sensing, and e-commerce.						

UNIT-I

Types of Security Attacks: Interruption, Interception, Modification, Fabrication, Security Services: Confidentiality, Authentication, Integrity, Non-repudiation, Access Control, Availability, Security Mechanisms, Model for Internetwork Security, Traditional/Classical Encryption Methods, Data Encryption Standard (DES) Algorithm, Security Strength of DES, Differential Cryptanalysis Technique, Linear Cryptanalysis Technique, Principles of Block Cipher Design, Block Cipher Modes of Operation, Blowfish Encryption Algorithm, Placement of Encryption in Communication Systems, Traffic Confidentiality in Networks, Cryptographic Key Distribution Methods, Random Number Generation in Cryptography.

UNIT-II

Principles of Public Key Cryptography, RSA Encryption Algorithm, Cryptographic Key Management Techniques, Diffie–Hellman Key Exchange Method, Elliptic Curve Cryptography (ECC), Message Authentication Concepts, Cryptographic Hash Functions, Authentication Requirements in Secure Communication, Authentication Functions, Message Authentication Techniques, Hash Functions and Message Authentication Codes (MACs), Hash and MAC Algorithms, SHA-512 Hash Algorithm, HMAC (Hash-based Message Authentication Code), IP Security Overview, IPsec Security Architecture, Authentication Header (AH), Encapsulating Security Payload (ESP), Combination of Security Associations, Key Management in IP Security.

UNIT-III

Introduction to Data Hiding, Data Hiding Models, Security and Privacy Issues in Data Hiding, Data Hiding Techniques in Digital Audio, Video, Images and Text, Basics of Steganography, Difference between Steganography and Cryptography, Classification of Steganography Techniques, Transform Domain Steganography, Spatial Domain Steganography, Statistical Steganography Methods, Other Steganography Techniques, Applications of Steganography, Covert Communication Channels, Steganography in Audio Data, Military Applications of Steganography, Steganography in E-commerce, Introduction to Digital Watermarking, Difference between Watermarking, Steganography and Cryptography, Watermarking Algorithms, Applications of Watermarking, Limitations of Watermarking.

UNIT-IV

Digital Rights Management Concepts, DRM Issues in E-commerce, Copyright Protection Techniques, Intellectual Property Protection in Digital Media, Security Issues in DRM Systems, Digital Signatures, Authentication Methods in Digital Systems, Case Studies in Digital Rights Management, Business Models for Digital Content Protection, Multimedia Security Concepts, Information Assurance in Multimedia Systems, Visual Cryptography Techniques, Cryptographic Key Management, Attacks on Data Hiding Systems, Performance Benchmarks for Data Hiding Techniques, Applications of Data Hiding in Medicine, Applications in Law Enforcement, Applications in Remote Sensing, Applications in E-commerce, Software Tools for Digital Data Hiding.

Reference Books:

1. Cryptography and Network Security (principles and approaches) by William Stallings Pearson Education, 4th Edition.
2. Ingemar Cox, Matthew Miller, Jeffrey Bloom, Jessica Fridrich, Ton Kalker, Digital Watermarking and Steganography, 2nd Edition, Morgan Kaufmann, 2007.
3. Michael T. Rago and Chet Hosmer, Data Hiding: Exposing Concealed Data in Multimedia, Operating Systems, Mobile Devices and Network Protocols, 1st Edition, Syngress, 2012
4. Network Security Essentials (Applications and Standards) by William Stallings Pearson Education.
5. Principles of Information Security, Whitman, Thomson.

B24-CS-CYS-OE-301	Soft Skills and Interpersonal Communication						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	0	0	3	70	30	100	3 Hours
Purpose	The course familiarizes students to perform effectively in job interviews and recruitment processes through improved communication and self-presentation skills..						
Course Outcomes (CO)							
CO1: Demonstrate effective verbal and non-verbal communication in professional situations.							
CO2: Apply interpersonal skills for teamwork, leadership and conflict management.							
CO3: Develop professional writing skills including emails, reports and job documents							
CO4: Deliver effective presentations using appropriate communication tools and techniques to demonstrate confidence in interviews, group discussions and workplace communication.							

Unit -1

Introduction to Soft Skills: Concept and importance of soft skills in engineering profession, Types of soft skills, Self-awareness and personality development, Emotional Intelligence, Positive attitude and motivation, Time management and goal setting.

Unit -2

Interpersonal Communication: Process and models of communication, Verbal and non-verbal communication, Barriers to communication and overcoming them, Listening skills and feedback techniques, Interpersonal relationships and empathy, Conflict management and negotiation skills.

Unit-3

Professional Communication: Principles of effective communication (7 Cs), Formal and informal communication, Business correspondence, Email writing and professional etiquette, Resume writing and cover letter, Technical report writing.

Unit-4

Presentation and Team Skills: Presentation skills and public speaking, Designing effective presentations using ICT tools, Group discussion techniques, Leadership and teamwork, Workplace etiquette and professional behavior, Cross-cultural communication.

Career Development Skills: Interview preparation techniques, Mock interviews and group discussion practice, Corporate communication and networking, Personal branding and professional ethics, Stress management and work-life balance.

Text Books

1. Meenakshi Raman & Sangeeta Sharma – Technical Communication: Principles and Practice. Oxford University Press.
2. Barun K. Mitra – Personality Development and Soft Skills. Oxford University Press.
3. K. Alex – Soft Skills: Know Yourself and Know the World. S. Chand Publishing.

Reference Books

1. Dale Carnegie – How to Win Friends and Influence People.
2. Herta Murphy – Effective Business Communication. McGraw Hill.
3. Lesikar & Flatley – Business Communication: Making Connections in a Digital World.
4. Stephen Covey – The 7 Habits of Highly Effective People.
5. Sanjay Kumar & Pushp Lata – Communication Skills. Oxford University Press.

B24-CS-CYS-OE-303	Innovation and Entrepreneurship						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal Assessment	Total	Time
3	-	-	3.0	70	30	100	3Hrs.
Purpose	To foster innovation-driven entrepreneurship for solving real-world challenges.						
Course Outcomes							
CO1	Explain the fundamental concepts and processes of innovation and Entrepreneurship.						
CO2	Identify entrepreneurial opportunities and generate innovative solutions for real-world problems.						
CO3	Analyze the startup ecosystem, incubation processes, and funding mechanisms for venture development.						
CO4	Understand the role of Intellectual Property Rights, including Patent, in protecting and commercializing innovations.						

UNIT-1

Fundamentals of Innovation: Concept and importance of innovation, Technological and non-technological innovation, Innovation process and stages, Idea generation techniques, Prototype development and design thinking, Current trends and developments in innovation

UNIT-2:

Entrepreneurship Overview: Meaning and scope of entrepreneurship, Entrepreneurial opportunities and identification, Entrepreneurial attitude, traits, and tendencies, social innovation and social entrepreneurship, Role of entrepreneurs in economic development. Dispelling Entrepreneurship Myths, Pros and Cons, Entrepreneurial Process.

UNIT-3:

Startup Ecosystem and Venture Development: Startup and venture development process, Innovation and startup ecosystem in India, Pre-incubation and incubation stages, Commercialization of innovations, Government schemes and funding support for startups

UNIT-4:

Mobilizing Resources: Acquiring and Utilizing Resources: Strategies for Effective Resource Management, Identifying and Assessing Financial Needs, Capital Estimation and Acquisition, Financing Sources such as Capital Markets, Angel Investors, and Venture Capital, Leveraging Mentorship and Information Channels.

Technology-driven Entrepreneurship: Navigating Market Dynamics: Recognizing Market Opportunities, Competitor Analysis, Brand Positioning, Intellectual Property Management, and Digital Marketing Strategies.

Suggested Books:

1. The Innovator's DNA: Mastering the Five Skills of Disruptive Innovators, Jeff Dyer, Hal Gregersen, C.M. Christensen, Harvard Business Review Press, 2011.
2. Design Thinking: Business Innovation, Maurício Vianna, Ysmar Vianna, Isabel K. Adler, Brenda Lucena, Beatriz Russo, MJV Press, 1st Electronic Edition, 2011 (also available at https://cdn2.hubspot.net/hubfs/1701231/Documents/Design_Thinking_-_The_Book/Design_Thinking_The_Book.pdf)
3. Contemporary Strategy Analysis: Text and Cases, Robert M Grant, Wiley, 9th Edition, 2016 (Chapter 6 and Chapter 9)
4. Timmons, J. and Spinelli, S. (2017). New Venture Creation: Entrepreneurship for the 21st Century. McGraw Hill Education
5. Shukla, M. (2020). Social Entrepreneurship in India: Quarter Idealism and a Pound of Pragmatism. Sage

- Publications. Bert C. and Lippitz Michael J. (2007), The Four Models of Corporate Entrepreneurship.
6. Hisrich, R.D., Manimala, M.J., Peters, M.P., Shepherd, D.A. (2017). Entrepreneurship, 9th Edition. McGraw Hill Education.

B24-CS-CYS-OE-305	Enterprise Resource Planning						
Lecture	Tutorial	Practical	Credits	Internal Assessment	End Semester Exam	Total Marks	Time
3	0	0	3	30	70	100	3hour
Purpose: To provide an understanding of concept of ERP and the ERP model;key terms; the transition from MRP to ERP; identify the levels of ERP maturity.							
Course Outcomes							
CO1	To Develop model for ERP for large projects						
CO2	To Develop model for E-commerce architecture for any application						
CO3	To Demonstrate a working knowledge of how data and transactions are integrated in an ERP system to manage the sales order process, production process, and procurement process.						
CO4	To Evaluate organizational opportunities and challenges in the design system within a business scenario.						

Unit I

ERP Introduction, Benefits, Origin, Evolution and Structure: Conceptual Model of ERP, the Evolution of ERP, the Structure of ERP.

Business Process Reengineering, Data ware Housing, Data Mining, Online Analytic Processing (OLAP), Product Life Cycle Management (PLM), LAP, Supply chain Management.

Unit II

ERP Marketplace and Marketplace Dynamics: Market Overview, Marketplace Dynamics, the Changing ERP Market. ERP- Functional Modules: Introduction, Functional Modules of ERP Software, Integration of ERP, Supply chain and Customer Relationship Applications.

Unit III

ERP Implementation Basics, ERP Implementation Life Cycle, Role of SDLC/SSAD, Object Oriented Architecture, Consultants, Vendors and Employees.

Unit IV

ERP & E-Commerce, Future Directives- in ERP, ERP and Internet, Critical success and failure factors, Integrating ERP into organizational culture. Using ERP tool: either SAP or ORACLE format to case study.

SUGGESTED BOOKS

- Vinod Kumar Garg and Venkitakrishnan N K,” Enterprise Resource Planning Concepts and Practice”, PHI.
- Joseph A Brady, Ellen F Monk, Bret Wagner,” Concepts in Enterprise Resource Planning”, Thompson Course Technology.
- Alexis Leon,” ERP Demystified”, Tata McGraw Hill
- Rahul V. Altekhar “Enterprise Resource Planning”, Tata McGraw Hill,
- Mary Summer, “Enterprise Resource Planning”- Pearson Education

B24-CS-CYS-301L	Computer Network Lab						
Lecture	Tutorial	Practical	Credit	Internal assessment	Practical Exam	Total	Time
0	0	3	1.5	40	60	100	3 Hours
Purpose	To explore networking concepts using Java programming & networking tools.						
Course Outcomes (CO)							
CO1	Do Problem Solving using algorithms.						
CO2	Design and test simple programs to implement networking concepts using Java.						
CO3	Document artifacts using applied addressing & quality standards.						
CO4	Design simple data transmission using networking concepts and implement.						

List of Practical

1. Create a socket for HTTP for web page upload and download.
2. Write a code simulating ARP /RARP protocols.
3. Study of TCP/UDP performance.
4. Performance comparison of MAC protocols
5. Performance comparison of routing protocols.
6. Write a program:
 - a. To implement echo server and client in java using TCP sockets.
 - b. To implement date server and client in java using TCP sockets.
 - c. To implement a chat server and client in java using TCP sockets.
7. Write a program:
 - a. To implement echo server and client in java using UDP sockets
 - b. To implement a chat server and client in java using UDP sockets.
 - c. To implement a DNS server and client in java using UDP sockets.
8. To flood the server from a spoofed source address leading to a DoS attack.
9. To sniff and parse packets that pass through using raw sockets.
10. To implement simple calculator and invoke arithmetic operations from a remote client.
11. To implement bubble sort and sort data using a remote client.
12. To simulate a sliding window protocol that uses Go Back N ARQ.

B24-CS-CYS-303L	Database Security Lab						
Lecture	Tutorial	Practical	Credits	Internal Assessment	Practical Exam	Total Marks	Time
0	0	3	1.5	40	60	100	3hour
Purpose: to understand how to protect digital assets from unauthorized access, misuse, and theft							
Course Outcomes							
CO1	To identify and analyze database security threats and vulnerabilities.						
CO2	To implement authentication, authorization, and access control mechanisms.						
CO3	To apply security policies and auditing techniques in DBMS.						
CO4	To design secure database applications.						

List of Practical

1. Install and configure a DBMS (MySQL / Oracle / PostgreSQL) and create a sample database environment.
2. Study fundamental data security requirements, security concerns, and possible risks in database systems.
3. Develop an enterprise database security policy specifying rules, procedures, and access control mechanisms.
4. Implement security enforcement techniques using constraints, views, and roles.
5. Perform vulnerability analysis of SQL queries to identify common database security issues.
6. Study database interaction and vulnerabilities in applications using Java/JDBC or PHP.
7. Demonstrate SQL injection attacks and implement methods to prevent them.
8. Implement user authentication mechanisms and secure login procedures.
9. Implement secure password storage techniques and enforce password management policies.
10. Create and manage database links and study their associated security risks.
11. Implement auditing mechanisms to track database access and user activities.
12. Design and implement a secure database system including authentication, access control, and auditing.

B24-CS-CYS-305L	Cryptography Lab						
Lecture	Tutorial	Practical	Credit	Internal assessment	Practical Exam	Total	Time
0	0	2	1	40	60	100	3 Hours
Purpose	To familiarize the students with the different cryptographic techniques.						
Course Outcomes (CO)							
CO1	To develop cryptography techniques using different cipher algorithms.						
CO2	To understand Number Theory and Algebra for design of cryptographic algorithms						
CO3	To understand the role of cryptography in communication over an insecure channel.						
CO4	To understand the symmetric key encryption schemes based on different security Models						

List of Practical

1. Write a program to perform encryption and decryption using the following algorithms:
a) Caesar Cipher b) Substitution Cipher c) Hill Cipher
2. Implementation of various Stream Ciphers, such as, additive cipher, multiplicative cipher, and affine cipher with cryptanalysis.
3. Write a program to implement the DES and AES algorithm logic
4. Implementation of Euclidean's algorithm, Chinese remainder theorem, Primality test, Fermat's algorithms, Euler algorithm.
5. Write a program to implement the Blowfish algorithm
6. Using Cryptography, encrypt the text "Hello world" using Blow Fish.
7. Create your own key using Java key tool.
8. Write a program to implement RSA Algorithm,
9. Implement the Diffie Hellman Key Exchange mechanism using HTML and JavaScript.
10. Configure SSH(SecureShell) and send/receive a file on this connection to verify the correctness of this system using the configured parameters.
11. Calculate the message digest of a text using the SHA,1 algorithm.
12. Calculate the message digest of a text using the MD5 algorithm.

B24-CS-CYS-302	Operating System						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3Hour
Purpose	To introduce the fundamental concepts and functions of operating systems including process management, memory management, file systems, and protection mechanisms. The course enables students to understand how operating systems manage hardware resources and provide services to application programs.						
Course Outcomes(CO)							
CO 1	Understand operating system concepts, structures, services, and process management including scheduling and threads.						
CO 2	Apply process synchronization techniques and analyze deadlock handling methods.						
CO 3	Explain memory management and I/O management techniques used in operating systems.						
CO 4	Understand file systems, access control, protection, and security mechanisms in operating systems.						

Unit-I

Computer system architecture and organization, Introduction and evolution of OS, Introduction to distributed OS, Real time systems and multimedia systems. OS structures: OS services, system calls and programs, OS design and implementation. Processes: Process concept, scheduling policies, algorithms, multilevel queuing, operations on process, Inter-process communication. Threads: multithreading models and threading issues. CPU scheduling: Criteria and algorithms, multiprocessor and thread scheduling.

Unit 2

Process synchronization: critical sections, classical two process and n-process solutions, hardware primitives for synchronization, semaphores, monitors, classical problems in synchronization (producer-consumer, readers-writer, dining philosophers, etc.). Deadlocks: modelling, resource allocation, characterization, prevention and avoidance, detection and recovery.

Unit 3

Memory management: Swapping, contiguous memory allocation, paging, multilevel paging, segmentation, demand paging, page replacement algorithms, allocation of frames, thrashing, working set model. Input/Output: I/O system and services, device controllers and device drivers, disks, scheduling algorithms and management.

Unit 4

File system interface: access methods, access control, directory structures, file organization, file sharing and protection. system performance, protection and security, OS design considerations for security, access control lists and OS support, internet and general network security. Operating system as service provider: Access control matrix, access control list, capability matrix, encryption and access permissions in Linux and Windows.

Reference Books:

1. A. Silberschatz, Peter B. Galvin and G. Gagne, "Operating System Concepts," (9th or newer edition), Wiley.
2. H. Brinch, "Operating System Principles," Prentice Hall of India.
3. Dhamdhere, "Systems programming & Operating systems," TataMcGrawHil
4. A.S. Tanenbaum, "Modern Operating Systems," Prentice Hall of India.

B24-CS-CYS-304	Digital Forensics						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3Hour
Purpose	To make students aware of cybercrimes and their modus operandi so they can analyze, investigate, and respond to digital attacks using forensic techniques.						
Course Outcomes(CO)							
CO 1	To understand the Fundamentals of Digital Forensics, Apply Digital Forensic Methodologies, Analyze Digital Evidence Using Forensic Tools.						
CO 2	To understand Windows Operating Systems for Forensics, Recover and Analyze Operating System Artifacts, Perform Practical Forensic Investigations on Windows Systems.						
CO 3	To understand Windows Operating Systems for Forensics, Recover and Analyze Operating System Artifacts, Conduct Forensic Investigations on Windows Systems.						
CO 4	To understand Legal and Regulatory Frameworks, Analyze Online and Electronic Evidence, Apply Legal and Ethical Standards in Investigations.						

Unit-1

Introduction to Digital Forensics: Overview, Comparison of Digital Forensics with other related areas, Evolution and historical development, Objectives and importance of Digital Forensics in cybercrime investigation, Stages and methodology involved in a Digital Forensic, Major domains and categories of Digital Forensics.

Overview of modern digital forensic tools: Requirements and categories of forensic tools, Comparison and evaluation of different forensic software tools, Computer forensic software utilities for data acquisition, Command-line&GUI based forensic tools, Hardware components used in investigations, Role and configuration of forensic workstations, Use and importance of writeblocking devices to protect digital evidence, Mobile device forensics and recovery of data from smartphones, Email and web browser forensics, Malware analysis and forensic investigation.

Unit-2

Identification and recognition of digital data: Time synchronization and timekeeping issues in digital evidence analysis, Detection and forensic examination of technical surveillance devices, Reconstruction of past digital events during investigations, Digital crime analysis, File formats and their role in forensic analysis, Handling corrupted or unsupported file formats, Techniques for file conversion and data recovery, Investigation of cybercrimes and network intrusion incidents, Network forensics, Monitoring of network traffic, Investigation of web-based attacks, Use of cyber forensic tools and case studies in digital investigations.

Overview of the Digital Forensics Investigation Process: Introduction to cybercrime scenes, Procedures for collecting digital evidence, Methods for preserving evidence, Techniques for forensic duplication and imaging, Investigation of live systems and analysis of dead systems. Application of hashing. Preparation and presentation of forensic investigation reports.

Unit-3

Overview of Windows operating systems in digital forensics: Study of Windows file system structures, Understanding the FAT filesystem, Analysis of the NTFS in forensic examinations, Comparison and summary of different Windows file systems, Examination of the Windows Registry for system and user activity information, Analysis of Windows Event Logs, Investigation of Prefetch files to identify executed programs, Examination of Shortcut (LNK) files, Forensic analysis of Windows executable files.

Overview of operating system artifacts: Techniques for recovering and analysing deleted data, Investigation of hibernation files, Detailed examination of Windows Registry artifacts, Analysis of Recycle Bin operations to track deleted file activities, Understanding and interpretation of file metadata in investigations, Use of system restore points for recovering, Analysis of shadow copies.

Unit-4

Overview of legal principles governing digital forensic investigations, Laws and regulations affecting the collection and analysis of digital evidence, Impact of legal requirements on digital forensic procedures, E-Discovery concepts in digital investigations, Case studies related to digital forensic investigations, Understanding internet-based resources used during forensic analysis, Forensic examination of web browsers and browsing artifacts, Analysis of email headers and email communication evidence, Investigation of activities on social networking platforms for digital evidence.

Reference Books:

1. Cory Altheide, Harlan Carvey, Digital Forensics with Open Source Tools, Syngress imprint of Elsevier.
2. Bill Nelson, Amelia Phillips, Christopher Steuart, "Guide to Computer Forensics and Investigations", Fourth Edition, Course Technology.
3. Angus M.Marshall, "Digital forensics: Digital evidence in criminal investigation", John – Wiley and Sons, 2008.
4. The Law of Evidence, Dr. Sr. Myneni, New Edition, Asian Law House, 2010

B24-CS-CYS-306	Ethical Hacking						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3Hour
Purpose	Ethical hackers simulate real-world attacks to uncover weaknesses in systems, networks, and applications.						
Course Outcomes(CO)							
CO 1	Organizations gain stronger defenses against modern cyber threats.						
CO 2	Early detection and remediation minimize chances of data theft or ransomware attacks.						
CO 3	Preventing breaches avoids financial losses, reputational damage, and regulatory fines.						
CO 4	Customers and stakeholders feel more confident in the organization’s ability to protect sensitive data.						

Unit-1

Introduction to Ethical Hacking: Definition and scope of ethical hacking, Types of hackers (white, black, grey hats), Legal and ethical considerations
 Footprinting and Reconnaissance: Information gathering techniques, Open-source intelligence (OSINT), Social engineering basics

Unit-2

Scanning and Enumeration: Network scanning techniques, Vulnerability scanning, Enumeration of services and users, Phishing, pretexting, baiting, Human factor in security,
 System Hacking: Password cracking methods, Privilege escalation, Malware and backdoors, Authentication bypass

Unit-3

Web Application Hacking, Common web vulnerabilities (SQL Injection, XSS, CSRF), OWASP Top 10, Wireless Network Hacking, Wireless standards and security, WPA/WPA2 vulnerabilities, Incident response lifecycle, Forensic process (collection, preservation, analysis)

Unit-4

Advanced Topics: Cloud security challenges, IoT vulnerabilities, Incident response and digital forensics, Real-world ethical hacking case studies, Best practices in penetration testing reports, Ethical hacking in organizations, Famous breaches (Equifax, Wanna Cry, Solar Winds)

Suggested books:

1. Hacking: The Art of Exploitation – *Jon Erickson*
2. Beginners Guide to Ethical Hacking & Cybersecurity – *Abhinav Ojha*
3. The Basics of Hacking and Penetration Testing – *Patrick Engebretson*

B24-CS-CYS-308	Cloud Computing						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	1	0	4	70	30	100	3Hour
Purpose	To introduce students to cloud computing concepts, virtualization, cloud services, and security practices so they can deploy, manage, and secure cloud-based systems in real-world scenarios.						
Course Outcomes(CO)							
CO 1	Understand the basic concepts of cloud computing, service models, and deployment models.						
CO 2	Explain virtualization technologies and cloud infrastructure components such as compute, storage, and networking.						
CO 3	Identify cloud security issues and apply security techniques such as identity management and data protection.						
CO 4	Deploy and manage basic cloud services and applications using cloud platforms.						

Unit-I

Fundamentals of Cloud Computing: Introduction to cloud computing, key characteristics and benefits, cloud deployment models: public, private, hybrid, cloud service models: IaaS, PaaS, SaaS, cloud ecosystem and study of NIST architecture.

Unit 2

Virtualization and Cloud Infrastructure: Concept of virtualization, benefits and applications. Types of virtualization: full virtualization, para-virtualization, and hardware-assisted virtualization. Hypervisors and virtual machines.

Introduction to Docker, Cloud architecture and cloud data centres, infrastructure management in cloud environments, Cloud services: compute, storage, and networking services, Scalability and load balancing techniques in cloud infrastructure, High availability and disaster recovery using virtualization.

Unit 3

Cloud Security: Security challenges in cloud environments, Identity and Access Management (IAM), Data encryption and privacy techniques, Threats, vulnerabilities, and compliance standards; Security best practices for cloud deployment.

Unit 4

Cloud Services & Application Deployment: Overview of AWS, Azure, and Google Cloud; Storage services, databases, and serverless computing
Monitoring, logging, and cost management; Deploying and securing cloud applications, Emerging trends: Edge computing, IoT, AI/ML on cloud.

Suggested Books:

1. "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, and Zaigham Mahmood.
2. "Cloud Computing: A Hands-On Approach" by ArshdeepBahga and Vijay Madiseti.
3. "Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" by Michael J. Kavis.
4. "Mastering Cloud Computing" by Rajkumar Buyya, Christian Vecchiola, and ThamaraiSelvi.

B24-BSC-CYS-302	Linear Algebra, and Statistics						
L	T	P	Credit	End Semester Exam	Internal Assessment	Total	Time
3	0	0	3	70	30	100	3 Hrs.
Purpose	To develop understanding of linear algebra, statistical inference, and regression analysis for solving mathematical and statistical problems in data analysis and research.						
CO 1	Understand and apply the concepts of vector spaces, basis, dimension, and linear transformations.						
CO 2	Analyze matrices using eigen values, characteristic polynomials, diagonalization, and decomposition theorems.						
CO3	Apply sampling theory and hypothesis testing methods to analyze statistical data.						
CO4	Use regression techniques to model relationships between variables and interpret statistical results.						

Unit 1

Vector spaces, subspaces, basis and dimension, coordinates, algebra of linear transformations, isomorphisms, representation of linear transformations by matrices, linear functionals, and annihilators, double dual and the transpose of a linear transformation.

Unit 2

Characteristic values and characteristic polynomials, diagonalizable transformations, annihilating polynomials, Cayley-Hamilton theorem, invariant subspaces, and triangular form, simultaneous triangularization and diagonalization, direct sum decompositions, invariant direct sums, primary decomposition theorem.

Unit 3

Definitions of random sample, parameter and statistic, sampling distribution of a statistic, sampling distribution of sample mean, standard errors of sample mean, sample variance and sample proportion. Null and alternative hypotheses, level of significance, Type I and Type II errors, their probabilities and critical region. Large sample tests, use of CLT for testing B.Sc. (Hons.) Statistics 33 single proportion, difference of two proportions, single mean, difference of two means, standard deviation and difference of standard deviations by classical and p-value approaches.

Unit 4

Linear Regression: Concept of regression; principle of least squares and fitting of straight line, derivation of two lines of regression, properties of regression coefficients, standard error of estimate obtained from regression line, correlation coefficient between observed and estimated values, distinction between correlation and regression. Angle between two lines of regression. Curvilinear Regression: Fitting of second degree parabola, power curve of the type $Y = axb$, exponential curves of the types $Y = abx$ and $Y = ae^{bx}$.

Suggested Books:

Text Books:

1. V. K. Rohatgi and A. K. Md. E. Saleh, An Introduction to Probability and Statistics, 2nd Edn., Wiley, 2001.
2. K. Hoffman and R. Kunze: Linear Algebra, 2nd Edition, Prentice Hall of India, 2005
3. S. Axler: Linear Algebra Done Right, 2nd Edition, Springer UTM, 1997.
4. Goon, A.M., Gupta, M.K., and B. Das Gupta: Fundamentals of Statistics, Vol-I. 2. Mukhopadhyaya, P.: Mathematical Statistics

Reference Books:

1. P. Halmos, Finite dimensional vectors paces, Springer, 1974.

2. Peter D. Lax, Linear algebra, Wiley student edition, 1997.
3. E.B. Vinberg, A course in algebra, Graduate text in Mathematics, volume 56, AMS, 2003.
4. M. Thamban Nair and Arindamasingh, Linear algebra, Springer, 2018
5. Hogg, R.V. and Tanis, E.A. (2009). A Brief Course in Mathematical Statistics. Pearson Education.
6. Johnson, R.A. and Bhattacharya, G.K. (2001). Statistics-Principles and Methods, 4th Ed., John Wiley and Sons.
7. Mood, M.A., Graybill, F.A. and Boes, C.D. (2007). Introduction to the Theory of Statistics, 3rd Ed., (Reprint).Tata McGraw-Hill Pub. Co. Ltd.

B24-CS-CYS-306L	Ethical Hacking Lab						
Lecture	Tutorial	Practical	Credit	Internal assessment	Practical Exam	Total	Time
0	0	3	1.5	40	60	100	3
Purpose	To simulate real-world attack scenarios in a controlled environment.						
Course Outcomes(CO)							
CO1	Set up and configure a secure virtual environment for penetration testing.						
CO2	Perform reconnaissance and scanning using tools like Nmap, Maltego, and Nessus.						
CO3	Identify and exploit vulnerabilities in operating systems, networks, and web applications.						
CO4	Conduct web application penetration testing using Burp Suite, SQLmap, and OWASP methodologies.						

List of Practical

1. Setting up a virtual lab environment (VMware/VirtualBox, Kali Linux, Metasploitable)
2. Practice basic **Linux commands** and scripting for security tasks.
3. Perform **WHOIS lookup** and DNS enumeration.
4. Vulnerability scanning with **Nessus/OpenVAS**.
5. Network mapping with **Zenmap**.
6. SQL Injection testing using **SQLmap**.
7. File upload vulnerability exploitation.
8. Wi-Fi password cracking using **Aircrack-ng**.
9. Simulated phishing attack using **SET (Social Engineering Toolkit)**.
10. Cloud penetration testing basics (AWS/Azure lab setup).

B24-CS-CYS-308L	Cloud Computing Lab						
L	T	P	Credit	Internal assessment	Practical Exam	Total	Time
0	0	3	1.5	40	60	100	3Hours
Purpose	The course is designed to provide practical knowledge of cloud computing concepts, virtualization, cloud services, and deployment of applications using cloud platforms.						
Course Outcomes							
CO1	To understand and perform basic operations on cloud platforms such as creating accounts, virtual machines, and storage services.						
CO2	To implement and manage cloud infrastructure including virtualization, containers, and networking services.						
CO3	To apply cloud security practices such as Identity and Access Management (IAM), encryption, and access control.						
CO4	To deploy and monitor cloud-based applications using cloud services and tools.						

List of Practical

1. To create and explore an account on a cloud platform such as AWS, Azure, or Google Cloud.
2. To create, start, stop, and manage a virtual machine in a cloud environment.
3. To create a cloud storage bucket and upload, download, and manage files.
4. To create users, groups, and roles and assign access permissions in the cloud.
5. To deploy and configure a basic web server on a cloud virtual machine.
6. To install Docker and run a basic containerized application.
7. To understand and implement basic data encryption for secure cloud storage.
8. To monitor CPU, storage, and network usage of cloud resources using cloud monitoring tools.
9. To understand scalability by configuring a load balancer for multiple cloud instances.
10. To deploy a simple cloud-based application integrating compute, storage, and security features.

B24-CS-CYS-304L	Digital Forensics Lab						
Lecture	Tutorial	Practical	Credit	Internal assessment	Practical Exam	Total	Time
0	0	2	1	40	60	100	3
Purpose/ PO	To provide hands-on experience in collecting, analyzing, and preserving digital evidence using various forensic tools, techniques, and methodologies across computers, networks, and online resources.						
Course Outcomes(CO)							
CO1	To Understand Digital Forensics Tools and Techniques, Identify and use various digital forensic tools to collect, preserve, and analyse digital evidence effectively.						
CO2	To Perform Network Forensic Analysis, Execute and interpret Windows and Linux network commands to gather forensic data and investigate network activities.						
CO3	To Recover and Analyse Deleted or Hidden DataApply techniques to recover deleted files and extract hidden information from images, audio files, or other media.						
CO4	To Apply Legal, Ethical, and Procedural Standards, handle digital evidence in compliance with legal and ethical guidelines while maintaining the integrity and admissibility of evidence.						

List of Practical

1. Introduction to Digital Forensics Tools:
Study the fundamentals of digital forensics and explore different forensic tools used for digital evidence collection, analysis, and investigation.
2. Network Investigation Commands in Windows:
Perform and analyse various Windows network commands such as ipconfig, ping, netstat, tracert, and nslookup for gathering network-related forensic information.
3. Network Investigation Commands in Linux:
Study and execute Linux networking commands such as ifconfig, ping, netstat, traceroute, whois, and nslookup to analyse network connectivity and gather forensic data.
4. Recovery of Deleted Files Using Forensic Tools:
Learn techniques to recover deleted files and folders using digital forensic software tools.
5. Data Hiding and Extraction Using Command Prompt:
Perform steganography techniques to hide a text file within an image or audio file and extract the hidden file using Command Prompt commands.
6. Creating a Forensic Image of a Hard Drive Using EnCase:
Study and perform the process of creating a forensic image (exact copy) of a hard drive using the EnCase Forensic tool.
7. Restoring Evidence from a Forensic Image Using EnCase:
Learn how to restore and examine digital evidence from a previously created forensic image file using EnCase.
8. Collection of Email Evidence from a Victim's Computer:
Study methods for identifying and collecting email-related evidence from a victim's PC during a digital forensic investigation.
9. File Comparison Using Compare It Software:

Perform forensic comparison of two files using Compare It software to detect differences and identify possible tampering.

10. Domain Information Gathering:

Collect and analyse information related to a given domain using tools and online resources for domain investigation.

11. Website Crawling and Banner Grabbing:

Perform information gathering by crawling websites and conducting banner grabbing using search engines such as Google.

12. Live Forensic Investigation Using Autopsy:

Conduct a live forensic investigation using the Autopsy digital forensics platform to analyse digital evidence.

B24-CS-CYS-OE-302	Law Of Privacy						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	-	-	3.0	70	30	100	3Hrs.
Purpose	The course familiarizes students with elementary knowledge of laws that would be of utility in their profession, including several new areas of law such as IPR, ADR						
Course Outcomes(CO)							
CO1	Introduce students to the concept and evolution of privacy rights in India and internationally.						
CO2	Understand constitutional, statutory, and technological aspects of privacy law						
CO3	Analyze the Right to Privacy as a Fundamental Right in India						
CO4	Study emerging issues such as data protection, surveillance, and digital privacy						

Unit -1

Concept and Evolution of Privacy: Meaning and concept of privacy, Historical development of privacy rights, Philosophical foundations of privacy, Privacy as a human right, International perspectives on privacy.

International Instruments: Universal Declaration of Human Rights (UDHR), International Covenant on Civil and Political Rights (ICCPR).

Unit -2

Constitutional Protection of Privacy in India: Right to Privacy under the Constitution of India, Article 21 and personal liberty, Judicial interpretation of privacy

Important Case Laws: Kharak Singh v. State of UP, Govind v. State of MP, Justice K.S. Puttaswamy v. Union of India (2017), R. Rajagopal v. State of Tamil Nadu

Unit-3

Data Protection and Informational Privacy: Concept of informational privacy, Personal data and sensitive personal data, Digital Personal Data Protection Act, 2023, Data protection principles, Role of Data Protection Authorities, Corporate responsibility and data security

Unit-4

Privacy in the Digital Age: Surveillance and state security, Privacy and social media, Privacy in cyberspace, Biometric data and Aadhaar, Artificial Intelligence and privacy concerns, Right to be forgotten

Books

1. Justice B.N. Srikrishna – *Data Protection in India*.
2. Graham Greenleaf – *Asian Data Privacy Laws*.
3. M.P. Jain – *Indian Constitutional Law*.
4. H.M. Seervai – *Constitutional Law of India*.
5. Avtar Singh – *Introduction to Cyber Law*.

Reference Books

1. Solove Daniel J. – *Understanding Privacy*.
2. Justice K.S. Puttaswamy Judgment (Supreme Court of India, 2017).
3. Apar Gupta – *Privacy Law and Technology*.
4. Lee A. Bygrave – *Data Privacy Law*.

B24-CS-CYS-OE-304	Reinforcement Learning						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal assessment	Total	Time
3	-	-	3.0	70	30	100	3Hrs.
Purpose	The purpose of this course is to provide students with a fundamental understanding of Reinforcement Learning (RL) and its core concepts. The course aims to help students learn how intelligent agents interact with environments, make decisions, and improve their performance through rewards and feedback. It also introduces exploration and exploitation strategies, Markov Decision Processes (MDP), and various value-based reinforcement learning algorithms used in modern AI applications.						
Course Outcomes(CO)							
CO1	Demonstrate various Components of Reinforcement Learning.						
CO2	Make use of various exploration and exploitation strategies.						
CO3	Apply Model based and Model Free Prediction techniques.						
CO4	Make use of different value based Reinforcement Learning Algorithms.						

UNIT-I

Introduction: Deep Reinforcement Learning, Suitability of RL, Components of Reinforcement Learning -Agent, Environment, Observations, Actions, Example-The Bandit Walk Environment, Agent-Environment interaction cycle, MDP (Markov Decision Process): The engine of the Environment-States, Actions, Transition Function, Reward.

UNIT-II

Planning: Objective of a decision making agent-environment, Plan, Optimal policy, Comparison of Policies, Bellman Equation/State-Value Function, Action-Value Function, Action-Advantage Function, Optimality. (ChExploitation and Exploration of Reinforcement Learning: Bandits- Single-state decision problem(Multi-Armed Bandit(MAB) problem), The cost of exploration, Approaches to solve MAB environments, Greedy Strategy, Random Strategy, Epsilon-Greedy Strategy, Decaying Epsilon-Greedy Strategy, Optimistic Initialization strategy, Strategic exploration, Softmax exploration strategy, Upper confidence bound (UCB) equation strategy, Thompson sampling strategy.

UNIT-III

Model Free Reinforcement Learning: Monte Carlo Prediction (MC), First-Visit MC (FVMC), Every-Visit MC (EVMC), Temporal Difference Learning (TD), Learning to estimate from multiple steps, N-step TD learning, Forward-view TD(λ), Backward-view TD(λ), Generalized policy iteration(GPI), Monte Carlo control, SARSA: On-Policy TD control, Q-learning: Off-Policy TD control, Double Q-learning, SARSA(λ), Watkins's Q(λ). Model Based Reinforcement Learning: Dyna-Q, Trajectory sampling.

UNIT-IV

Policy Based Reinforcement Learning: Policy Gradient and Actor-Critic Methods—REINFORCE Algorithm and Stochastic Policy Search, Vanilla Policy Gradient(VPG), Asynchronous Advantage Actor-Critic (A3C), Generalized Advantage Estimation (GAE), Advantage Actor-Critic(A2C), Deep Deterministic Policy Gradient (DDPG), Twin-Delayed DDPG (TD3), Soft Actor-Critic (SAC), proximal policy optimization (PPO).

Suggested Book:

1. Miguel Morales, Grokking Deep Reinforcement Learning, Manning Publications.
2. Richard S. Sutton and Andrew G. Barto, Reinforcement learning: An Introduction, Second Edition, MIT Press, 2019.
3. Marco Wiering, Martijn van Otterlo(Ed), Reinforcement Learning, State-of-the-Art, Adaptation, Learning, and Optimization book series, ALO, volume 12, Springer.
4. Keng, Wah Loon, Graesser, Laura, Foundations of Deep Reinforcement Learning: Theory and Practice in

Python, Addison Wesley Data & Analytics Series, 2020.

5. Francois Chollet, Deep Learning with Python, Manning Publications.

B24-CS-CYS-OE-306	RESEARCH METHODOLOGY						
Lecture	Tutorial	Practical	Credit	End Semester Exam	Internal Assessment	Total	Exam Time
3	0	0	3	70	30	100	3 Hour
Purpose: To understand in the systematic, rigorous investigation of digital threats, vulnerabilities, and defense mechanisms							
Course Outcomes (CO)							
CO 1	Develop an understanding of how to identify research topics						
CO 2	Conduct literature reviews and understand the research process.						
CO 3	Develop skills in collecting and analyzing qualitative/quantitative data.						
CO 4	To draft research reports/papers/proposals properly and adhere to research ethics and intellectual property rights						

UNIT-I

Definition and objective of research, types of research, steps in research process, research Design: concept and types, defining and formulating the research problems, importance of literature review- primary and secondary sources, reviews, research databases, web sources, identifying gap areas from the literature and research database, surveying synthesis, Interpretation.

UNIT-II

Primary and Secondary data; data collection techniques: observation, interview, questionnaires, schedules, case study; levels of measurement; problems in measurement in research – validity, reliability. Descriptive statistics: mean, median, mode, range, mean deviation and standard deviation, regression and correlation analysis, inferential statistics: t- tests, Chi-square tests. Correlation (rank difference and product moment), Analysis of variance (ANOVA) (one way)

UNIT-III

Types of research reports, steps of writing a report, layout of a report, layout of a research paper, ethical issues related to publishing, plagiarism, and self-plagiarism. Conflict of research, Predatory publishers and Journals, Open access publication, citation and acknowledgement, software tools for similarity check

UNIT-IV

Patents: process of filing a patent at the national and international levels; introduction and significance of intellectual property rights; commercialization, royalties, copyright; trade-related aspects of IPR; administration of the patent system in India; licensing and transfer of technology; case studies.

Suggested Books:

1. Walpole R.A., Myers R.H., Myers S.L. and Ye, King: Probability & Statistics for Engineers and Scientists, Pearson Prentice Hall, Pearson Education, Inc. 2007.
2. Kothari C.K., Research Methodology- Methods and Techniques (New Age International, New Delhi), 2004.
3. English for Writing Research Papers by Adrian Wallwork, 2nd Edition. Springer
4. N.S. Gopalakrishnan & T.G. Agitha, Principles of Intellectual Property (2009), Eastern Book Company, Lucknow.
5. D G Montgomery, Design and analysis of Experiments, John Wiley India Edition